

Entry trajectory optimization considering blackout zone communication constraint[☆]

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Abstract

The communication between the reusable launch vehicle (RLV) and the ground station will be shielded by the blackout during the entry flight. However, the communication utilizing the tracking and data relay satellite (TDRS) may be feasible. Therefore, a blackout zone communication (BZC) constraint for three-dimensional trajectory is proposed to stabilize the RLV-TDRS communication. First, the angle between the RLV's antenna pointing and the RLV-TDRS vector is expressed, and the communication constraint between the RLV and TDRS (or other satellites) is formulated. Second, the BZC constraint is completed by combining with the altitude range of the blackout. Third, considering the BZC constraint, the trajectory optimization problem is established as a two-phase optimal control problem. Simulations across three scenarios illustrate the effectiveness of the method. Compared with trajectories without the BZC constraint, the proposed method can restrict the antenna pointing within the specified range.

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Keywords: Reusable launch vehicle; Tracking and data relay satellite; Blackout communication; Trajectory optimization

1. Introduction

Reusable launch vehicles (RLVs) transport crew and payloads between space and the Earth, which is crucial for the sustainable use of space resources (Mishra and Sushnigdha, 2024). Therefore, several RLVs, including Space Shuttle Orbiter, X-37B, X-33, and China's reusable experimental spacecraft, have been developed over the last few decades (Harpold and Gavert, 1983; Hollis et al., 2001). Because an RLV enters the atmosphere at hypersonic speeds, the blackout occurs for a certain period (Takahashi, 2015; Wu et al., 2018).

When entering the blackout zone, the RLV is surrounded by high-temperature plasma, which is caused by the free electrons and ions generated during flight (Hartunian et al., 2007; Ramjatan et al., 2017). Due to the influence of the plasma, communication between RLV and ground station cannot be maintained (Makarov et al., 2005). In addition, the radio signal of the global positioning system is blocked during the blackout. Since the inertial measuring unit becomes the sole source of navigation information, the navigation errors will accumulate throughout the blackout (Wu et al., 2021). Because this phenomenon may last over more than 10 min and the RLV flies at high Mach number, significant state dispersion may occur (Wang et al., 2023).

However, communication by utilizing the tracking and data relay satellite (TDRS) may be feasible if the RLV satisfies certain constraints. The TDRS has been widely used to provide communication services, including telemetry, tracking and control (TT&C) and data transmission for

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low and middle earth orbit spacecrafts (such as Space Shuttle and International Space Station), rockets and other aircraft (Flaherty et al., 2006). To guarantee stable communication between the vehicle and the TDRS, the pointing of the vehicle's antenna relative to the RLV-TDRS vector must be restricted within a specified range. This paper refers to this requirement on the antenna pointing as the “RLV-TDRS communication constraint”.

If the antenna's pointing satisfies the communication constraint during the blackout zone, the RLV-TDRS communication can be regarded as feasible and stable. Note that this RLV-TDRS communication constraint is not a constraint applicable to the entire flight. Once the RLV has exited the blackout zone, it is no longer necessary to consider the constraint. Therefore, this RLV-TDRS communication constraint is referred to as the “blackout zone communication (BZC) constraint”.

This paper assumes that the RLV-TDRS communication during the blackout zone is achieved via a TDRS in a geosynchronous orbit with a low inclination angle. With the RLV's total flight time of only 800–2000 s, the TDRS can be considered stationary relative to the Earth. Using the TDRS's position and the RLV's state, the angle between the vehicle's antenna pointing and the RLV-TDRS vector is derived. This angle is referred to as the “relative angle of antenna pointing (RAAP)”. Once the RAAP is calculated, the RLV-TDRS communication constraint can be derived according to the beam width of the vehicle's antenna, and the BZC constraint can be obtained. Finally, the trajectory optimization problem considering the BZC constraint is formulated as a two-phase optimal control problem (OCP).

With the development of numerical optimization, various methods have emerged to solve entry trajectory optimization problems (Malyuta et al., 2021), such as the pseudospectral method (Garg et al., 2010; Zhang et al., 2023), sequential convex programming (SCP) (Bae et al., 2024; Liu et al., 2016), and heuristic algorithms (Su et al., 2021). For more complex constraints, Zhao and Song (2017) used the multi-phase convex programming to handle the waypoint constraints during entry flight. Zhang and Cui (2023) utilized the sequential mixed-integer linear programming to solve the optimal trajectory with general-shape no-fly zone constraints. Compared with these sequential iteration methods, the pseudospectral method can directly handle nonlinear constraints (Darby et al., 2011; Garg et al., 2010). Since the proposed BZC constraint is a highly nonlinear and complex constraint, the pseudospectral method is employed to solve the established two-phase OCP.

The main contributions of this paper are as follows.

- (1) The RAAP and the RLV-TDRS communication constraint for three-dimensional trajectory is formulated.
- (2) The BZC constraint is proposed, combining the RLV-TDRS communication constraint with the blackout's altitude range.
- (3) The trajectory optimization problem considering BZC constraint is established as a two-phase OCP.

The rest of this paper is organized as follows. Section 2 introduces the RLV-TDRS communication constraint and derives its expression. Section 3 proposes the BZC constraint and formulates the trajectory optimization problem as a two-phase OCP. Section 4 analyzes the optimal trajectories considering the BZC constraint through simulations.

2. RLV-TDRS communication constraints

Denote by $\mathbf{P}$ and $\mathbf{S}$ the positions of the RLV and the TDRS, respectively. Denote by $\mathbf{A}$ and $\mathbf{W}$ the vehicle's antenna pointing, and the vector from the RLV to the TDRS, respectively. The vector $\mathbf{W}$ can be calculated by

$$\mathbf{W} = \mathbf{S} - \mathbf{P} \quad (1)$$

Denote by β the RAAP, which is the relative angle between $\mathbf{W}$ and $\mathbf{A}$. According to the definition, β should satisfy

$$\cos \beta = \frac{\mathbf{A} \mathbf{W}}{\|\mathbf{A}\| \|\mathbf{W}\|} \quad (2)$$

The RLV-TDRS communication constraint on the antenna pointing is defined as

$$\beta \leq \eta/2 \quad (3)$$

where η represents the beam width of the vehicle's antenna. The diagram of this constraint is depicted in Fig. 1.

The coordinate systems used to calculate the value of β are defined in Table 1. Denote by $O - X_T Y_T Z_T$ the Half-Velocity Coordinate System, $O - X_B Y_B Z_B$ the Body Coordinate System, $O_C - X_O Y_O Z_O$ the Geocentric Coordinate System, and $O_C - X_E Y_E Z_E$ the North Up East (NUE) Coordinate System, respectively.

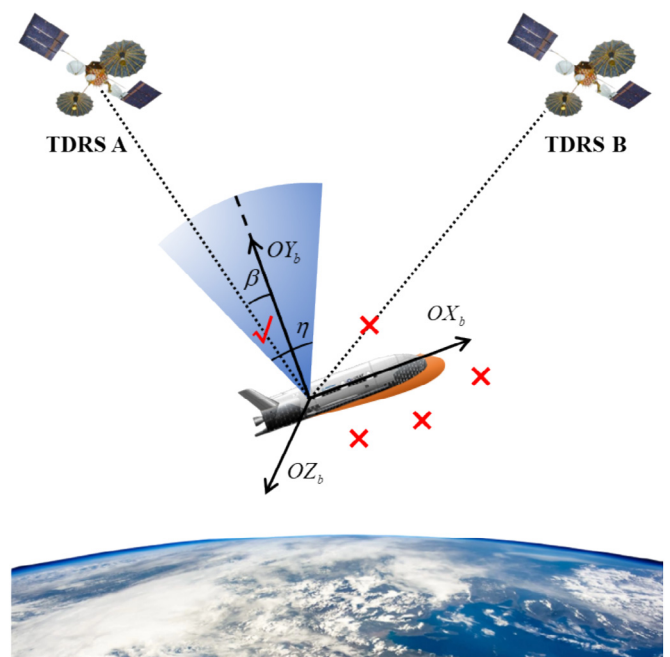


Fig. 1. Diagram of the RLV-TDRS communication constraint.

Table 1
Definitions of the coordinate systems.

Coordinate system	Definition
Body Coordinate System $O - X_B Y_B Z_B$	O : the RLV's center of mass OX_B : point to the RLV's body nose OY_B : perpendicular to OX_B in the RLV's longitudinal symmetry plane (up) OZ_B : perpendicular to $X_B Y_B$ (right)
Half-Velocity Coordinate System $O - X_T Y_T Z_T$	OX_T : aligned with the velocity OY_T : perpendicular to OX_T , upward in local vertical plane OZ_T : perpendicular to $X_T Y_T$ (right)
NUE Coordinate System $O - X_E Y_E Z_E$	Fixed geographic directions (NUE)
Geocentric Coordinate System $O_C - X_O Y_O Z_O$	O_C : the Earth's center $O_C X_O$: along the Earth's spin axis, pointing to the North Pole $O_C Y_O$: point to the equator in the prime meridian plane $O_C Z_O$: perpendicular to $X_T Y_T$ (right)

Because the plasma density varies significantly with the location on the vehicle's surface, the antenna's position is critical to the RLV-TDRS communication. For example, the plasma density at the vehicle's rear is significantly lower than at its abdomen. Therefore, we assume that the antenna's pointing aligns with OY_B , and the expression of $\mathbf{A}$ in the $O - X_B Y_B Z_B$ is

$$\begin{bmatrix} A_{xB} \\ A_{yB} \\ A_{zB} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (4)$$

where A_{xB} , A_{yB} , and A_{zB} represent the coordinates of $\mathbf{A}$ in $O - X_B Y_B Z_B$.

The expressions of $\mathbf{P}$ and $\mathbf{S}$ in $O_C - X_O Y_O Z_O$ are

$$\begin{bmatrix} P_{xO} \\ P_{yO} \\ P_{zO} \end{bmatrix} = \begin{bmatrix} r \sin \phi \\ r \cos \phi \cos \theta \\ r \cos \phi \sin \theta \end{bmatrix} \quad (5)$$

$$\begin{bmatrix} S_{xO} \\ S_{yO} \\ S_{zO} \end{bmatrix} = \begin{bmatrix} r_s \sin \phi_s \\ r_s \cos \phi_s \cos \theta_s \\ r_s \cos \phi_s \sin \theta_s \end{bmatrix} \quad (6)$$

where P_{xO} , P_{yO} , and P_{zO} represent the coordinates of $\mathbf{P}$ in $O_C - X_O Y_O Z_O$, respectively, S_{xO} , S_{yO} , and S_{zO} represent the coordinates of $\mathbf{S}$ in $O_C - X_O Y_O Z_O$, respectively, r , θ , and ϕ are the distance from the Earth's center to the RLV, the RLV's longitude, and latitude, respectively, r_s , θ_s , and ϕ_s are the distance from the Earth's center to the satellite, the satellite's longitude, and latitude, respectively.

Substitute Eqs. (5) and (6) into Eq. (1), we have

$$\begin{bmatrix} W_{xO} \\ W_{yO} \\ W_{zO} \end{bmatrix} = \begin{bmatrix} S_{xO} \\ S_{yO} \\ S_{zO} \end{bmatrix} - \begin{bmatrix} P_{xO} \\ P_{yO} \\ P_{zO} \end{bmatrix} \quad (7)$$

The expression of $\mathbf{W}$ in $O_C - X_E Y_E Z_E$ can be calculated by

$$\begin{bmatrix} W_{xE} \\ W_{yE} \\ W_{zE} \end{bmatrix} = \mathbf{G}_{OE} \begin{bmatrix} W_{xO} \\ W_{yO} \\ W_{zO} \end{bmatrix} \quad (8)$$

where W_{xE} , W_{yE} , and W_{zE} represent the coordinates of $\mathbf{W}$ in $O_C - X_E Y_E Z_E$, respectively. The matrix $\mathbf{G}_{OE}$ can be calculated by

$$\mathbf{G}_{OE} = \begin{bmatrix} \cos(-\phi) & \sin(-\phi) & 0 \\ -\sin(-\phi) & \cos(-\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad (9)$$

The expression of $\mathbf{W}$ in $O - X_T Y_T Z_T$ can be calculated by

$$\begin{bmatrix} W_{xT} \\ W_{yT} \\ W_{zT} \end{bmatrix} = \mathbf{G}_{ET} \begin{bmatrix} W_{xE} \\ W_{yE} \\ W_{zE} \end{bmatrix} \quad (10)$$

where W_{xT} , W_{yT} , and W_{zT} represent the coordinates of $\mathbf{W}$ in $O - X_T Y_T Z_T$, respectively. Denote by γ and ψ the RLV's flight-path angle and heading angle, respectively. The matrix $\mathbf{G}_{ET}$ can be calculated by

$$\mathbf{G}_{ET} = \begin{bmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-\psi) & 0 & -\sin(-\psi) \\ 0 & 1 & 0 \\ \sin(-\psi) & 0 & \cos(-\psi) \end{bmatrix} \quad (11)$$

The expression of $\mathbf{W}$ in $O - X_B Y_B Z_B$ can be calculated by

$$\begin{bmatrix} W_{xB} \\ W_{yB} \\ W_{zB} \end{bmatrix} = \mathbf{G}_{TB} \begin{bmatrix} W_{xT} \\ W_{yT} \\ W_{zT} \end{bmatrix} \quad (12)$$

where W_{xB} , W_{yB} , and W_{zB} represent the coordinates of $\mathbf{W}$ in $O - X_B Y_B Z_B$, respectively. The matrix $\mathbf{G}_{TB}$ can be calculated by

$$\mathbf{G}_{TB} = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \sigma & \sin \sigma \\ 0 & -\sin \sigma & \cos \sigma \end{bmatrix} \quad (13)$$

Reorganizing Eqs. (9), (11), and (13), we have

$$\begin{bmatrix} W_{xB} \\ W_{yB} \\ W_{zB} \end{bmatrix} = \mathbf{G}_{TB} \mathbf{G}_{ET} \mathbf{G}_{OE} \begin{bmatrix} W_{xO} \\ W_{yO} \\ W_{zO} \end{bmatrix} \quad (14)$$

Once the coordinates of $\mathbf{W}$ is obtained by Eq. (14), the value of β can be calculated by Eq. (2) and used to guarantee the RLV-TDRS communication constraint in Eq. (3).

3. Problem formulation

3.1. Dynamics

Let v , L , D , σ , and Ω denote the dimensionless speed, lift acceleration, drag acceleration, bank angle, and the Earth self-rotation rate, respectively. The dimensionless dynamics for the entry flight of a RLV is expressed as (Zhang et al., 2025):

$$\dot{r} = v \sin \gamma \quad (15)$$

$$\dot{\theta} = v \cos \gamma \sin \psi / (r \cos \phi) \quad (16)$$

$$\dot{\phi} = v \cos \gamma \cos \psi / r \quad (17)$$

$$\dot{v} = -D - \frac{\sin \gamma}{r^2} + p_v \quad (18)$$

$$\dot{\gamma} = \frac{L}{v} \cos \sigma - \left(\frac{1}{r} - v^2 \right) \frac{\cos \gamma}{vr} + p_\gamma \quad (19)$$

$$\dot{\psi} = \frac{L}{v \cos \gamma} \sin \sigma + \frac{v}{r} \cos \gamma \sin \psi \tan \phi + p_\psi \quad (20)$$

where

$$p_v = \Omega^2 r \cos \phi (\sin \gamma \cos \phi - \cos \gamma \sin \phi \cos \psi) \quad (21)$$

$$p_\gamma = 2\Omega \cos \phi \sin \psi + \frac{\Omega^2 r \cos \phi (\cos \gamma \cos \phi + \sin \gamma \sin \phi \cos \psi)}{v} \quad (22)$$

$$p_\psi = -2\Omega (\tan \gamma \cos \phi \cos \psi - \sin \phi) + \frac{\Omega^2 r \sin \phi \cos \phi \sin \psi}{v \cos \gamma} \quad (23)$$

represent the accelerations caused by the Earth self-rotation. The altitude of the vehicle can be calculated by $h = R_0(r - 1)$, where h is the altitude, and $R_0 = 6378$ km is the Earth's radius.

The lift and drag accelerations can be calculated by

$$L = R_0 C_L \rho v^2 S_{ref} / (2m) \quad (24)$$

$$D = R_0 C_D \rho v^2 S_{ref} / (2m) \quad (25)$$

where ρ is the atmospheric density, m is the vehicle's mass, S_{ref} , C_L , and C_D are the reference area, lift coefficient, and drag coefficient, respectively. In this paper, a conceptual RLV is used (Lu, 1997), whose C_L and C_D are determined by the angle of attack (AoA). The AoA varies along a prescribed profile with respect to the speed, as shown in Fig. 2.

Denote by α the AoA. C_L and C_D can be calculated by

$$C_L = c_{10} + c_{11}\alpha + c_{12}\alpha^2 \quad (26)$$

$$C_D = c_{20} + c_{21}C_L + c_{22}C_L^2 \quad (27)$$

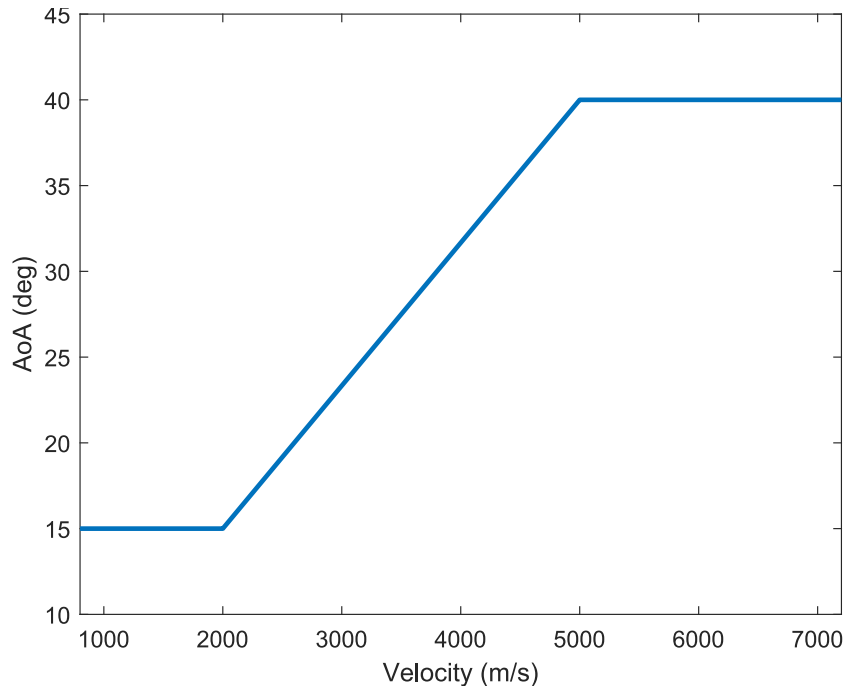


Fig. 2. The AoA profile.

where $c_{10} = -0.041065$, $c_{11} = 0.016292$, $c_{12} = 0.0002602$, $c_{20} = 0.080505$, $c_{21} = -0.03026$, $c_{22} = 0.86495$.

3.2. Constraints

Since the AoA is prescribed as a function of the speed, the trajectory is determined by the bank angle once the initial states are specified. The magnitude of the bank angle should satisfy

$$|\sigma| \leq \sigma_{\max} \quad (28)$$

where $\sigma_{\max}$ represents the upper bound of the bank angle's magnitude. The change rate of the bank angle should satisfy

$$|\dot{\sigma}| \leq \omega_{\max} \quad (29)$$

where $\omega_{\max}$ represents the upper bound of the bank angle rate's magnitude.

The dynamic pressure, load factor, and heating rate can be calculated by

$$q = 0.5R_0g_0\rho v^2 \quad (30)$$

$$n = \sqrt{L^2 + D^2} \quad (31)$$

$$\dot{Q} = k_q\sqrt{\rho}v^{3.15} \quad (32)$$

where q , n , and $\dot{Q}$ represent the dynamic pressure, load factor, and heating rate, respectively. They should satisfy

$$q \leq q_{\max}, n \leq n_{\max}, \dot{Q} \leq \dot{Q}_{\max} \quad (33)$$

where $q_{\max}$, $n_{\max}$, and $\dot{Q}_{\max}$ represent the upper bounds of the corresponding quantities, respectively.

The diagram of the BZC constraint is displayed in Fig. 3. The vehicle should satisfy Eq. (3) in the blackout zone to ensure stable RLV-TDRS communication. Once

the vehicle exits the blackout zone, the RLV-TDRS communication constraint is no longer required.

The RLV's initial altitude is regarded as the blackout zone's upper altitude bound. Although the RLV may exceed the upper bound, the duration is sufficiently short and thus the communication constraint is still considered. Therefore, the BZC constraint is expressed as

$$\begin{aligned} \beta &\leq \eta/2, h \geq h_{BZ} \\ \beta &\text{ is free, } h < h_{BZ} \end{aligned} \quad (34)$$

where h_{BZ} represents the lower altitude bound of the blackout zone.

Denote the state vector by $\mathbf{x} = [r \theta \phi v \gamma \psi]^T$. The initial state variables are specified, i.e.,

$$\mathbf{x}(t_0) = \mathbf{x}_0 \quad (35)$$

where t_0 is the initial time, and $\mathbf{x}_0 = [r_0 \theta_0 \phi_0 v_0 \gamma_0 \psi_0]^T$ is the initial state vector. The terminal states should satisfy

$$\mathbf{x}_f - \delta\mathbf{x}_f \leq \mathbf{x}(t_f) \leq \mathbf{x}_f + \delta\mathbf{x}_f \quad (36)$$

where t_f is the terminal time, $\mathbf{x}_f = [r_f \theta_f \phi_f v_f \gamma_f \psi_f]^T$ represents the desired terminal state vector, and $\delta\mathbf{x}_f$ represents the allowable error of the terminal state.

3.3. Multi-phase optimal control problem

To limit the change rate of the bank angle, σ is regarded as an augmented state variable, and the bank angle rate $\dot{\sigma}$ is used as the control variable, i.e.,

$$u = \dot{\sigma} \quad (37)$$

where u represents the control variable.

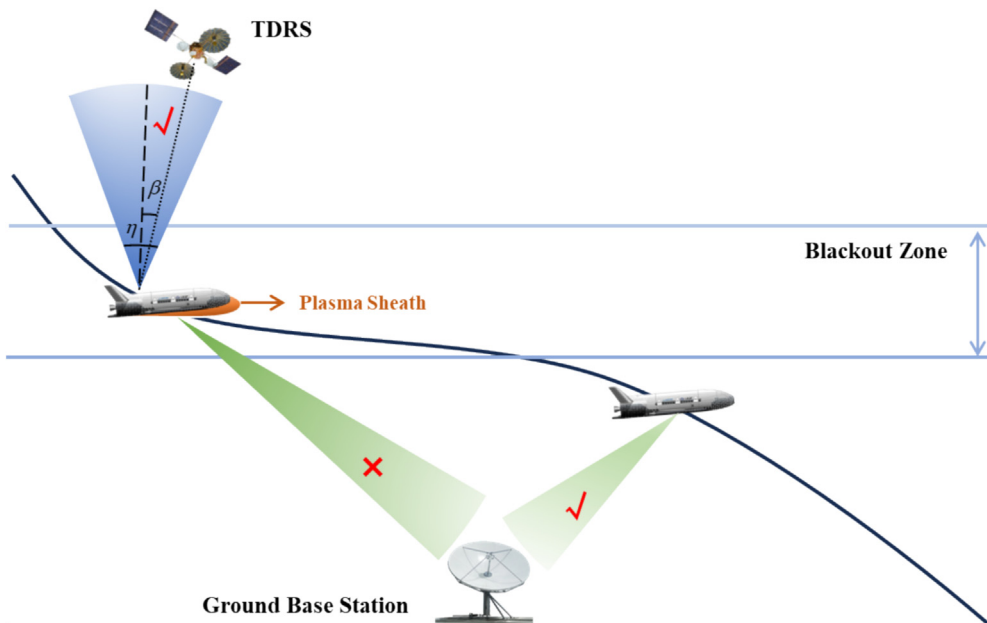


Fig. 3. Diagram of the BZC constraint.

Denote by $\tilde{\mathbf{x}} = [r \theta \phi v \gamma \psi \sigma]^T$ the augmented state vector, $\mathbf{f}$ the augmented dynamics. Eqs. (15)–(20) and Eq. (37) can be reformulated as

$$\dot{\tilde{\mathbf{x}}} = \mathbf{f}(\tilde{\mathbf{x}}, u) \quad (38)$$

To keep the bank angle as smooth as possible, the performance index is defined as

$$J = \min \int_{t_0}^{t_f} |u| dt \quad (39)$$

Subsequently, the trajectory optimization problem considering BZC constraint can be formulated as the following optimal control problem

$$\begin{aligned} \text{P0: } \min \quad & J = \int_{t_0}^{t_f} |u| dt \\ \text{s.t.} \quad & \dot{\tilde{\mathbf{x}}} = \mathbf{f}(\tilde{\mathbf{x}}, u) \\ & |\sigma| \leq \sigma_{\max}, |u| \leq \omega_{\max} \\ & q \leq q_{\max}, \dot{Q} \leq \dot{Q}_{\max}, n_y \leq n_{\max} \\ & \begin{cases} \beta \leq \eta/2, h \geq \underline{h}_{BZ} \\ \beta \text{ is free}, h < \underline{h}_{BZ} \end{cases} \\ & \mathbf{x}(t_0) - \mathbf{x}_0 = 0, -\delta \mathbf{x}_f \leq \mathbf{x}(t_f) - \mathbf{x}_f \leq \delta \mathbf{x}_f \end{aligned} \quad (40)$$

The BZC constraint is not considered when $h < \underline{h}_{BZ}$. However, this needs to determine whether the RLV's altitude is higher than $\underline{h}_{BZ}$. To simplify the problem, we give the following assumption.

Assumption 1. The altitude being equal to $\underline{h}_{BZ}$ occurs only once.

Remark. Although RLV's altitude may skip a few times, it usually happens when the altitude is significantly higher than $\underline{h}_{BZ}$. Therefore, assumption 1 holds for most cases.

Based on assumption 1, the trajectory can be divided into two phases according to $\underline{h}_{BZ}$, and the BZC constraint only needs to be considered in the first phase. Hence, problem P0 can be transformed into the following two-phase OCP:

$$\begin{aligned} \text{P1: } \min \quad & J = \int_{t_0}^{t_m} |u_{(1)}| dt + \int_{t_m}^{t_f} |u_{(2)}| dt \\ \text{s.t.} \quad & \dot{\tilde{\mathbf{x}}}_{(i)} = \mathbf{f}(\tilde{\mathbf{x}}_{(i)}, u_{(i)}) \\ & |\sigma_{(i)}| \leq \sigma_{\max}, |u_{(i)}| \leq \omega_{\max} \\ & q_{(i)} \leq q_{\max}, \dot{Q}_{(i)} \leq \dot{Q}_{\max}, n_{y(i)} \leq n_{\max}, i = 1, 2 \\ & \beta(\tilde{\mathbf{x}}_{(1)}) \leq \eta/2, h_{(1)} \geq \underline{h}_{BZ}, h_{(2)} \leq \underline{h}_{BZ} \\ & \tilde{\mathbf{x}}_{(2)}(t_m) - \tilde{\mathbf{x}}_{(1)}(t_m) = 0 \\ & \mathbf{x}_{(1)}(t_0) - \mathbf{x}_0 = 0, -\delta \mathbf{x}_f \leq \mathbf{x}_{(2)}(t_f) - \mathbf{x}_f \leq \delta \mathbf{x}_f \end{aligned} \quad (41)$$

where t_m is the switching time from the first phase to the second phase, and the subscript (i) represents the i th phase of the trajectory.

The constraints $h_{(1)} \geq \underline{h}_{BZ}$, $h_{(2)} \leq \underline{h}_{BZ}$, and $\tilde{\mathbf{x}}_{(2)}(t_m) - \tilde{\mathbf{x}}_{(1)}(t_m) = 0$ imply the equation $h_{(1)}(t_m) = \underline{h}_{BZ}$, which may increase the solution difficulty. Therefore, we hope to remove this equality constraint. By removing the constraint $h_{(1)} \geq \underline{h}_{BZ}$, problem P1 is transformed into the following form:

$$\begin{aligned} \text{P2: } \min \quad & J = \int_{t_0}^{t_m} |u_{(1)}| dt + \int_{t_m}^{t_f} |u_{(2)}| dt \\ \text{s.t.} \quad & \dot{\tilde{\mathbf{x}}}_{(i)} = \mathbf{f}(\tilde{\mathbf{x}}_{(i)}, u_{(i)}) \\ & |\sigma_{(i)}| \leq \sigma_{\max}, |u_{(i)}| \leq \omega_{\max} \\ & q_{(i)} \leq q_{\max}, \dot{Q}_{(i)} \leq \dot{Q}_{\max}, n_{y(i)} \leq n_{\max}, i = 1, 2 \\ & \beta(\tilde{\mathbf{x}}_{(1)}) \leq \eta/2, h_{(2)} \leq \underline{h}_{BZ} \\ & \tilde{\mathbf{x}}_{(2)}(t_m) - \tilde{\mathbf{x}}_{(1)}(t_m) = 0 \\ & \mathbf{x}_{(1)}(t_0) - \mathbf{x}_0 = 0, -\delta \mathbf{x}_f \leq \mathbf{x}_{(2)}(t_f) - \mathbf{x}_f \leq \delta \mathbf{x}_f \end{aligned} \quad (42)$$

where t_m is the switching time from the first phase to the second phase.

Assumption 2. The optimal trajectories of problem P1 and P2 exist and are unique.

The only difference between problems P1 and P2 is that P2 does not consider the inequality constraint $h_{(1)} \geq \underline{h}_{BZ}$. This means that the terminal altitude of the first phase may lower than $\underline{h}_{BZ}$. Since the BZC constraint is considered throughout the first phase, it needs to be satisfied in some period outside the blackout zone. Although it seems to be stricter, the optimal trajectories of these two problems are equivalent.

Proposition 1. The optimal trajectories of problem P1 and P2 are equivalent.

Proof. Denote the optimal solutions of problems P1 and P2 by $(u_1^*, \tilde{\mathbf{x}}_1^*, t_m^*)$ and $(u_2^*, \tilde{\mathbf{x}}_2^*, \hat{t}_m^*)$, respectively. Denote the optimal performance indices of problem P1 and P2 by J_1^* and J_2^* , respectively. Obviously, $(u_1^*, \tilde{\mathbf{x}}_1^*, t_m^*)$ is a feasible solution of problem P2. Hence, there must be $J_1^* \geq J_2^*$.

Because $h_{(2)} \leq \underline{h}_{BZ}$, there is $h_{2(1)}^*(\hat{t}_m^*) \leq \underline{h}_{BZ}$, where the subscript “2” represents problem P2. According to assumption 1, there must exist a moment $\hat{t}_a^* \in (t_0, \hat{t}_m^*)$ which satisfies the following conditions

$$h_2^*(\hat{t}_a^*) = \underline{h}_{BZ} \quad (43)$$

$$h_2^*(\tau) > \underline{h}_{BZ}, \tau \in [t_0, \hat{t}_a^*) \quad (44)$$

$$h_2^*(\tau) < \underline{h}_{BZ}, \tau \in (\hat{t}_a^*, t_f] \quad (45)$$

According to the definition of problem P2, there must be

$$\beta(\tilde{\mathbf{x}}_2^*(\tau)) \leq \eta/2, \tau \in [t_0, \hat{t}_a^*] \quad (46)$$

Define a pair of switching time, control and state as follow

$$\begin{aligned}
t'_m &= \hat{t}_a^* \\
u'_{(1)} &= u_2^*(\tau), \tau \in [t_0, \hat{t}_a^*] \\
\tilde{\mathbf{x}}'_{(1)} &= \tilde{\mathbf{x}}_2^*(\tau), \tau \in [t_0, \hat{t}_a^*] \\
u'_{(2)} &= u_2^*(\tau), \tau \in [\hat{t}_a^*, t_f] \\
\tilde{\mathbf{x}}'_{(2)} &= \tilde{\mathbf{x}}_2^*(\tau), \tau \in [\hat{t}_a^*, t_f]
\end{aligned} \quad (47)$$

Substituting Eq. (47) into problem P1, the following conditions are satisfied

$$\begin{aligned}
\beta(\tilde{\mathbf{x}}'_{(1)}) &\leq \eta/2 \\
h'_{(1)} &\geq \underline{h}_{BZ} \\
h'_{(2)} &\leq \underline{h}_{BZ}
\end{aligned} \quad (48)$$

This indicates that Eq. (47) is a feasible solution of problem P1. Denote by $J_1^\#$ the performance index. Obviously, there is $J_1^\# = J_2^*$.

Since $(u_1^*, \tilde{\mathbf{x}}_1^*, t_m^*)$ is the optimal solution of problem P1, there must be $J_1^\# = J_2^* \geq J_1^*$. Because $J_1^* \geq J_2^*$, the condition $J_1^* = J_2^*$ must hold. According to assumption 2, the optimal trajectories of problem P2 and P1 are equivalent.

4. Simulations

Three different scenarios are displayed in Table 2. The initial altitude of these three scenarios is 80 km, which is regarded as the upper altitude bound of the blackout zone. Other initial states are displayed in Table 2. The longitude and latitude of the target are 111.3 deg and 42.2 deg, respectively. The terminal altitude and velocity are 25 km and 800 m/s, respectively. The allowable error of the terminal longitude, latitude, altitude, and velocity are 0.5 deg, 0.5 deg, 0.5 km, and 60 m/s, respectively. The lower altitude bound of the blackout zone is 60 km. Other constraints and parameters are displayed in Table 3. The trajectories are solved by using the software GPOPS-II (Patterson and Rao, 2016).

4.1. Different scenarios

The altitude-velocity profiles and ground tracks of these scenarios are depicted in Fig. 4. The desired terminal state constraints are satisfied by all of the trajectories. Note that the altitude-velocity profiles, which reflect the change process of the vehicle's mechanical energy, vary significantly different over these scenarios. This is because that, apart from the initial altitude, the other initial states of these sce-

narios are different. Specifically, because scenario 3 has the highest initial speed and a zero flight-path angle, while those of other scenarios are negative, there is a significant “skip” during the initial period. Although the altitude exceeds the initial altitude, which is regarded as the upper bound of the blackout zone, the BZC constraint is still considered because this interval is sufficiently short.

The RAAP of scenarios 1–3 are depicted in Fig. 5. All RAAPs satisfy the BZC constraint during the blackout zone. In contrast, because the RLV can communicate directly with the ground station outside the blackout zone, the RAAP is allowed to exceed the RLV-TDRS communication constraint. Therefore, the BZC constraint can effectively restrict the antenna pointing within a suitable range.

The bank angle and its change rate are depicted in Fig. 6. The BZC constraint is satisfied by adjusting the bank angle through planning the bank angle rate. Since the performance index aims to control the bank angle as smooth as possible, the bank angle rates remain at a small magnitude, and the bank angle histories vary smoothly during most periods of the entry flight. Taking scenario 1 as an example, the bank angle remains constant during some periods, and the transition between different periods are gradual.

Fig. 7 illustrates the flight-path angle of scenarios 1–3. The flight-path angle in the first phase may be positive for short intervals. In particular, that of scenario 3 remains positive during the initial stage, so the altitude exceeds the initial altitude. However, the flight-path angle remains negative throughout the second phase, maintaining the altitude below $\underline{h}_{BZ}$. Therefore, transforming the original OCP into the two-phase OCP is feasible.

The path constraints histories are depicted in Fig. 8. All of the path constraints are satisfied in these three scenarios.

Table 3
Parameters of the simulations.

Parameter	Value	Unit
$q_{\max}$	20	kPa
$n_{\max}$	2.5	/
$\dot{Q}_{\max}$	160,000	W
$\sigma_{\max}$	60	deg
$\omega_{\max}$	10	deg/s
$\eta/2$	45	deg
$\underline{h}_{BZ}$	60	km
λ_s	77	deg
ϕ_s	0	deg
r_s	42,164	km

Table 2
Initial states of different scenarios.

Scenario	Longitude (deg)	Latitude (deg)	Velocity (m/s)	Flight-path angle (deg)	Heading angle (deg)
1	55	25	6800	−0.5	45
2	65	10	7000	−1	50
3	75	0	7200	0	50

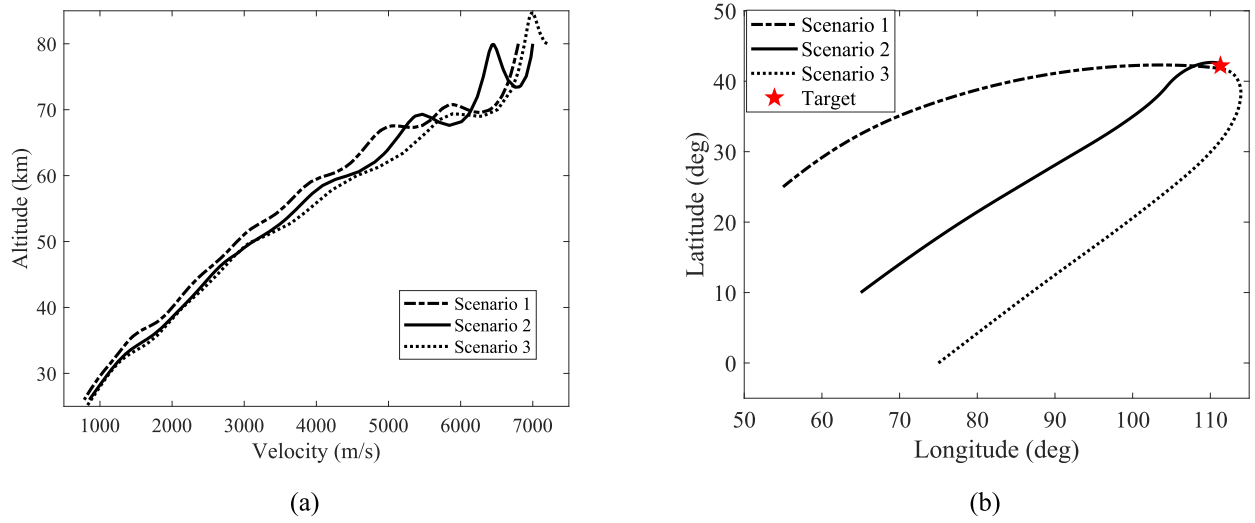


Fig. 4. Altitude-velocity profiles and ground tracks of scenarios 1–3 considering the BZC constraint. (a) Altitude-velocity profiles. (b) Ground tracks (longitude-latitude profiles).

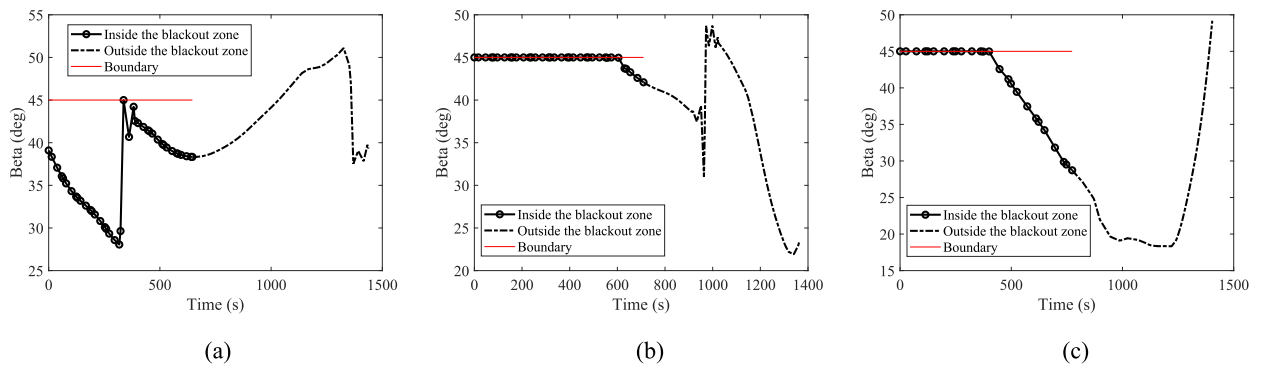


Fig. 5. RAAP histories of scenarios 1–3 considering the BZC constraint. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3.

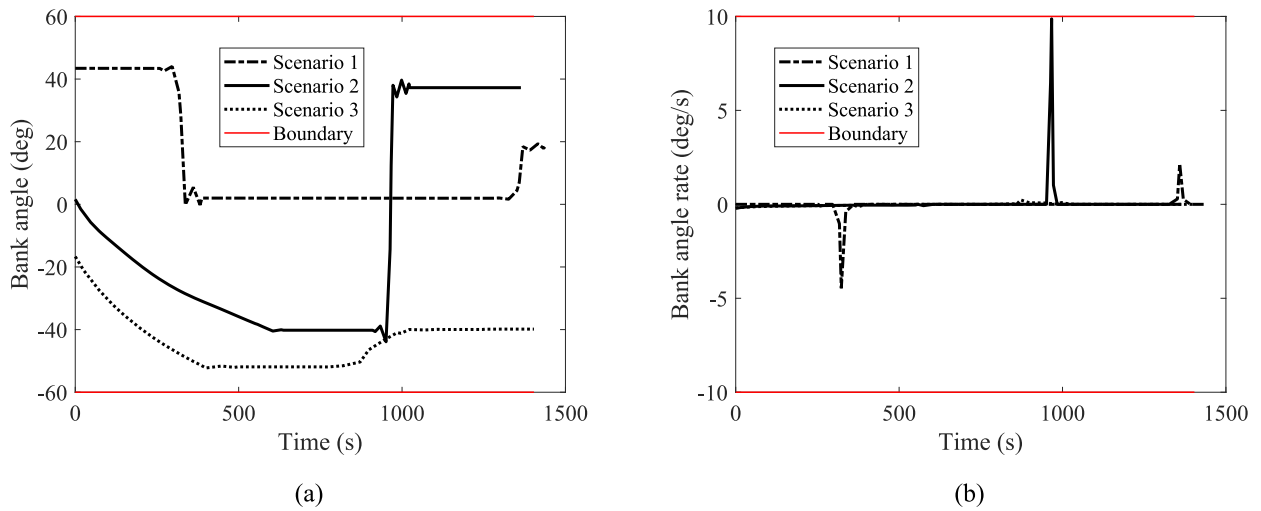


Fig. 6. Bank angle and its change rate of scenarios 1–3 considering the BZC constraint. (a) Bank angle histories. (b) Bank angle rate histories.

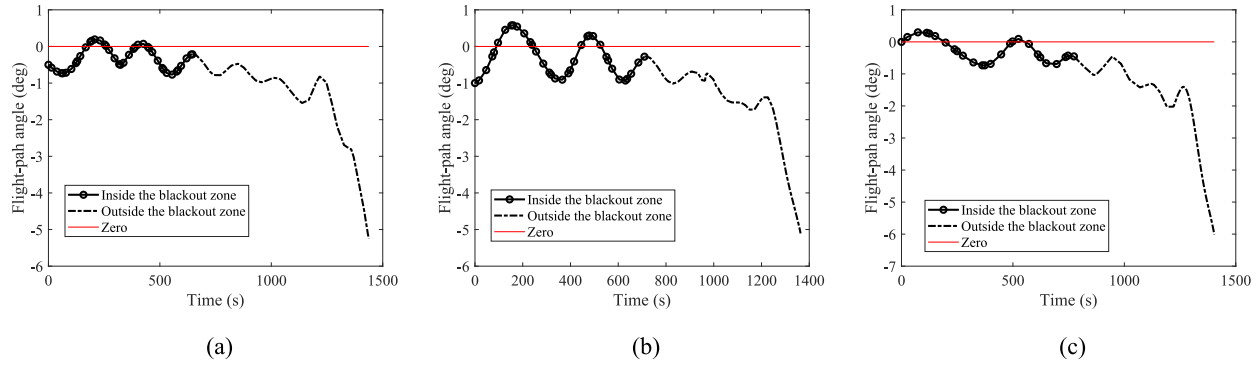


Fig. 7. Flight-path angle histories of scenarios 1–3 considering the BZC constraint. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3.

Therefore, the RAAP, which is a complex quantity coupling different state and control variables, can meet the BZC constraint under the premise that other path constraints are satisfied.

4.2. Comparisons

Figs. 9 and 10 illustrate the comparisons of the RAAP and the bank angle with and without BZC constraint. In

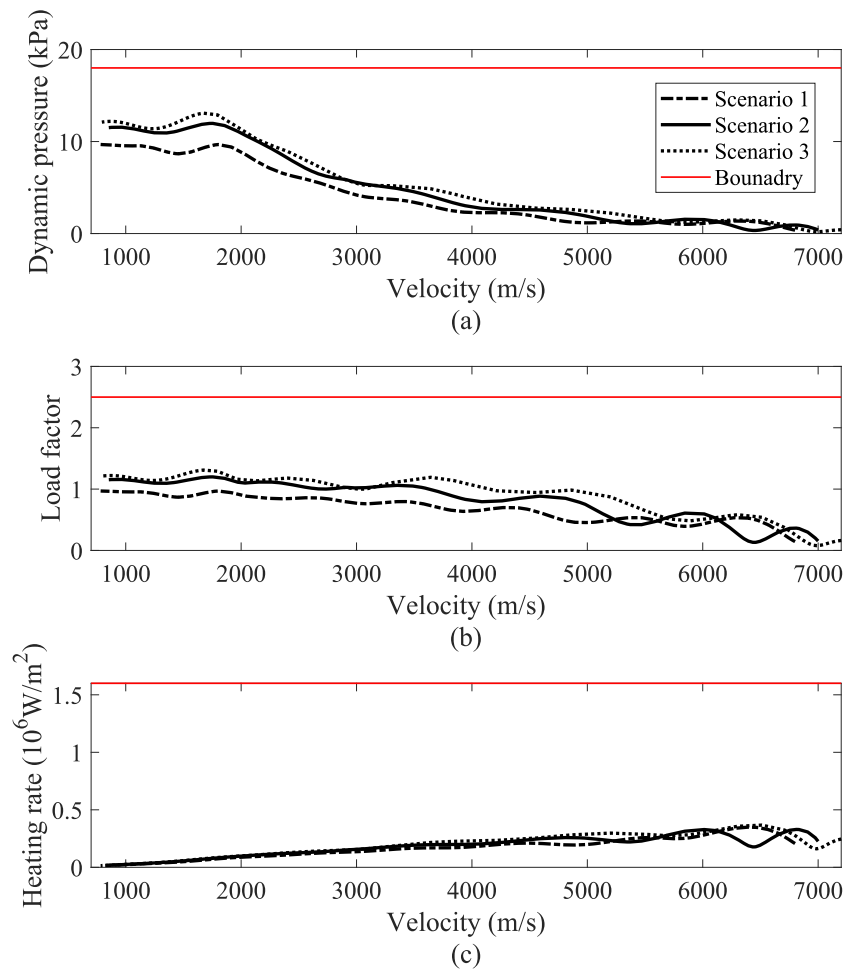


Fig. 8. Path constraint histories of scenarios 1–3 considering the BZC constraint. (a) Dynamic pressure histories. (b) Load factor histories. (c) Heating rate histories.

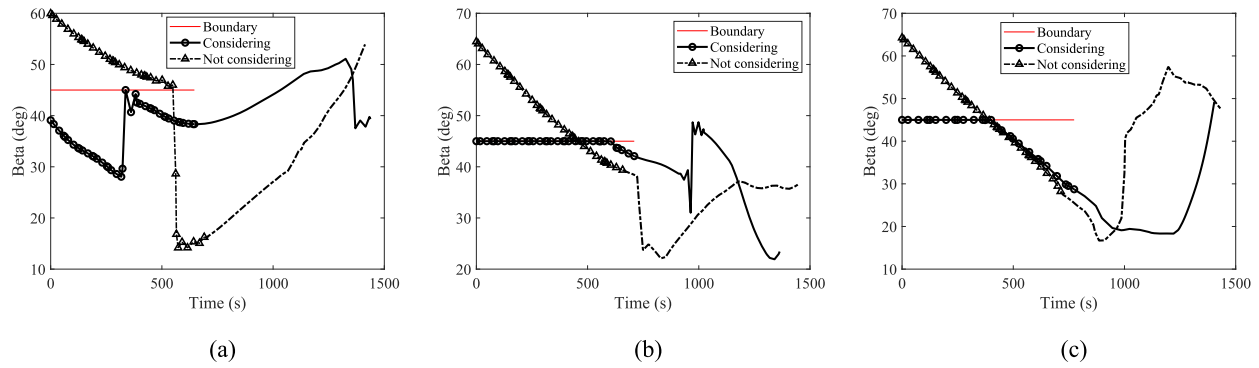


Fig. 9. Comparisons of RAAP histories among considering and NOT considering BZC constraint. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3.

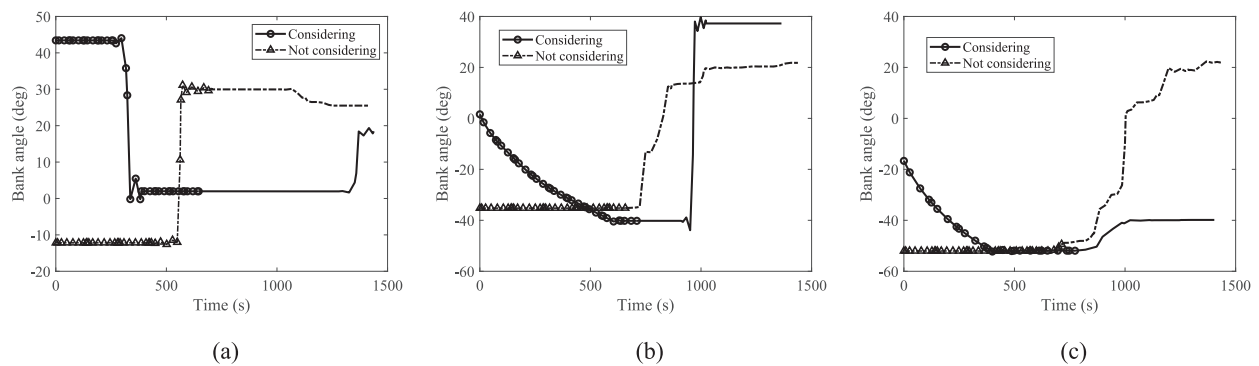


Fig. 10. Comparisons of bank angle histories among considering and NOT considering the BZC constraint. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3.

all these scenarios, the RAAPs of the optimal trajectory without BZC constraint, which is obtained by using the same initial guess and solution method, exceeds the bound during some intervals when the RLV is in the blackout zone. As a result, the RLV-TDRS communication will be disrupted if the RLV flies along these trajectories. In con-

trast, the RAAPs considering the BZC constraint remain below the upper bound during the blackout zone, strongly indicating the validity of the BZC constraint.

Fig. 10 illustrates the bank angle histories of scenarios 1–3. Since the BZC constraint is guaranteed by adjusting the bank angle, there exist remarkable distinctions between

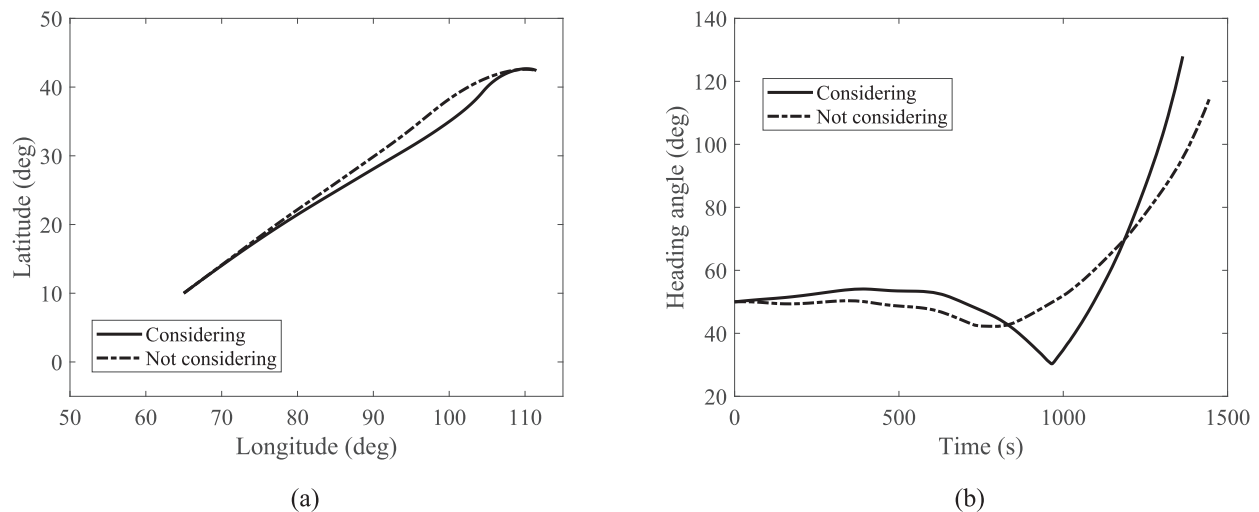


Fig. 11. Comparisons of heading angle and ground tracks among considering and NOT considering the BZC constraint in scenario 2. (a) Ground tracks. (b) Heading angle histories.

the bank angle histories with and without considering the constraint. Although the bank angle without the constraint is more stable, both of them change smoothly.

The lateral state variables, which consist of the longitude, latitude and heading angle, are also significantly different. This is caused by the different variation processes of the bank angle. Using scenario 2 as an example, the ground tracks and heading angle are depicted in Fig. 11. To maintain the RLV-TDRS communication during the blackout, the bank angle considering BZC constraint is positive at the initial stage, causing a slight increase in the heading angle. This indicates that the BZC constraint may have significant influence on the lateral profile.

5. Conclusions

The blackout can last over 10 min during the RLV's entry flight. During this period, the RLV cannot communicate directly with the ground station, but can potentially communicate with the TDRS. To maintain stable RLV-TDRS communication, the three-dimensional trajectory must keep the RAAP, which represents the angle between the vehicle's antenna pointing and the RLV-TDRS vector, within a specified range. Therefore, the BZC constraint is defined as the constraint on the RAAP when the vehicle is in the blackout zone. The trajectory optimization problem considering BZC constraint is established as a two-phase optimal control problem. Simulations validate the effectiveness of the above strategy. The optimal trajectory accounting for the BZC constraint keeps the RAAP within the required range during the blackout zone, providing favorable conditions for RLV-TDRS communication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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